

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

Unit 4: Data Center Networks

1

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Slide adopted from Profs. Alizadeh, Jain, and Zhangs' materials

Agenda

- Modern Data Centers
- Design Goals of Data Center Networks
- (Old) Data Center Networks
- Fat-Tree Network Topology
- Virtual Layer 2 (VL2) Networks

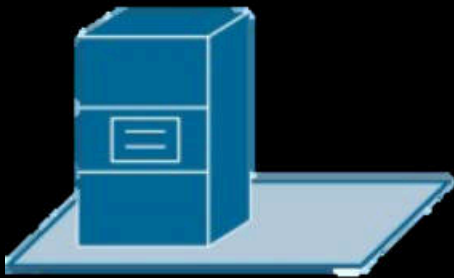
What is Data Center

- Large facilities with 10s of thousands of networked servers
 - Compute, storage, and networking working in concert
 - Warehouse-Scale Computers
 - Huge investment: ~ 0.5 billion for large datacenter



Data Center Servers

4



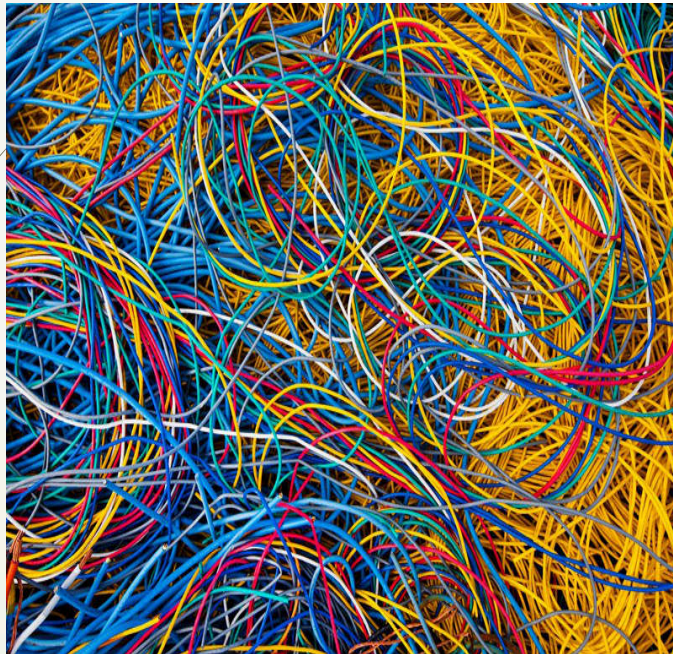
They Are Expensive

Amortized Cost*	Component	Sub-Components
~45%	Servers	CPU, memory, disk
~25%	Power infrastructure	UPS, cooling, power distribution
~15%	Power draw	Electrical utility costs
~15%	Network	Switches, links, transit

- Server cost dominates
- Networking cost nontrivial
- Long provisioning timescale: Quarterly purchases in the best scenario
 - Careful design is a Must

Cabling

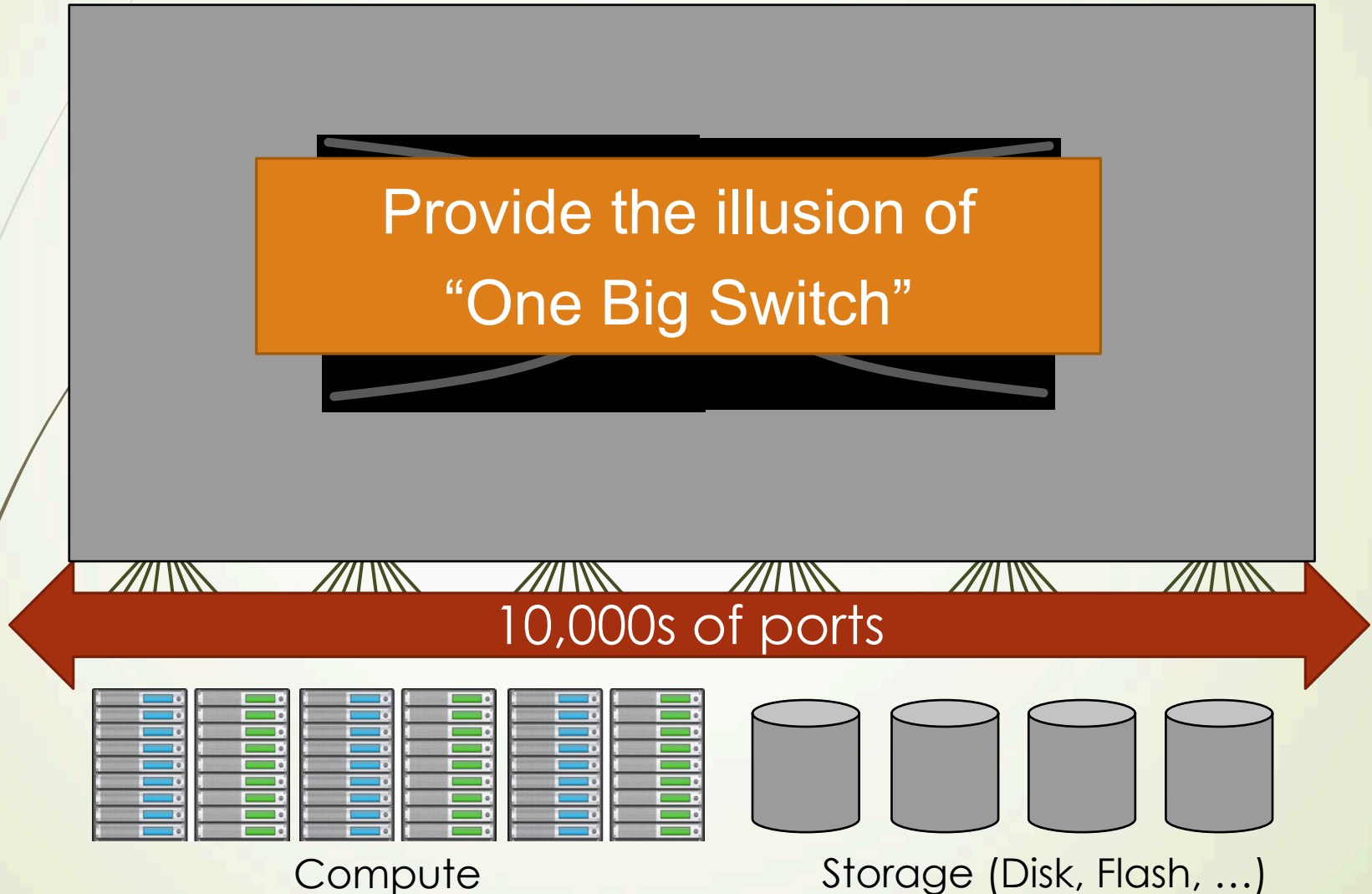
Unstructured



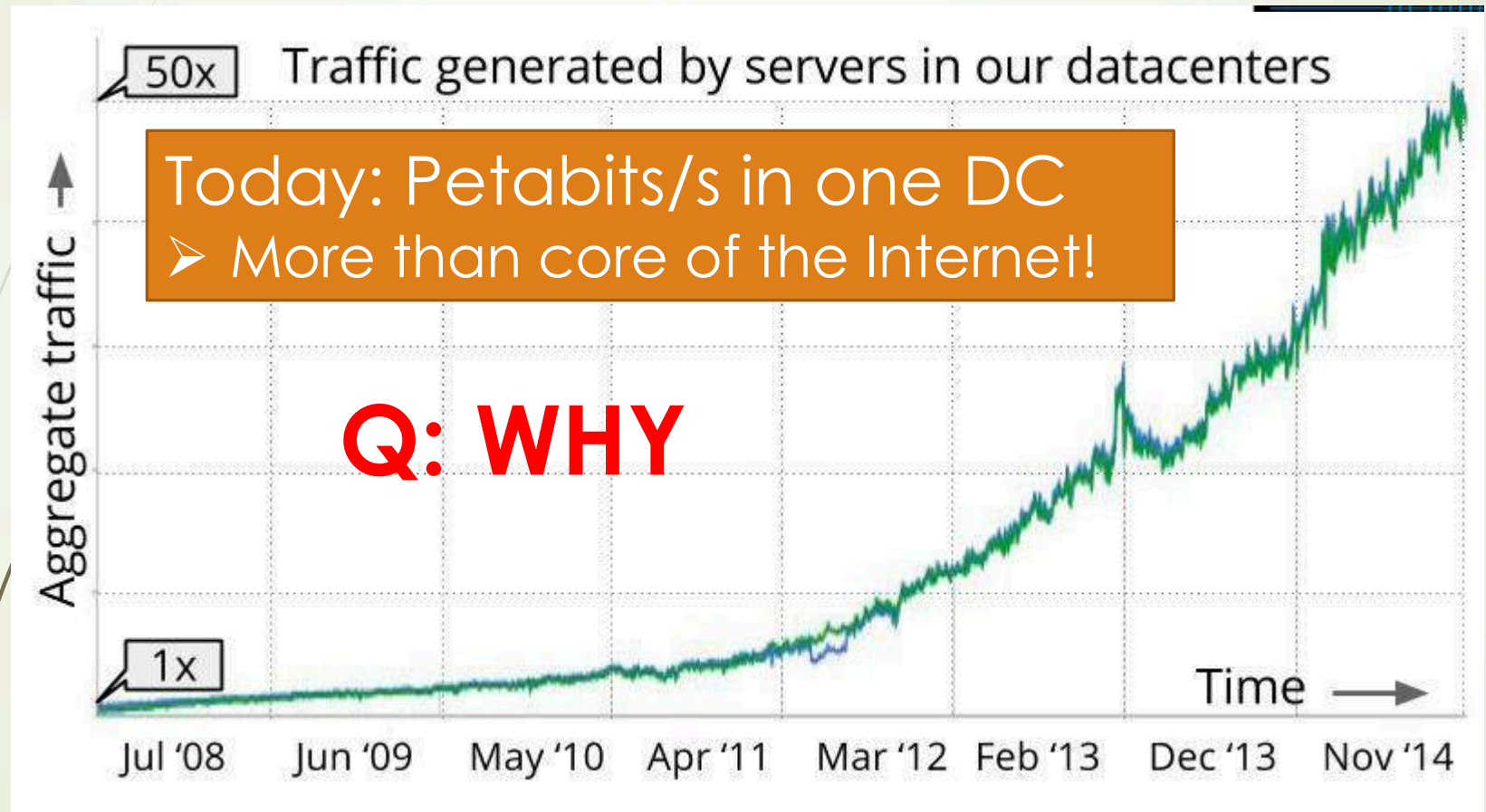
Structured



Data Center Networks



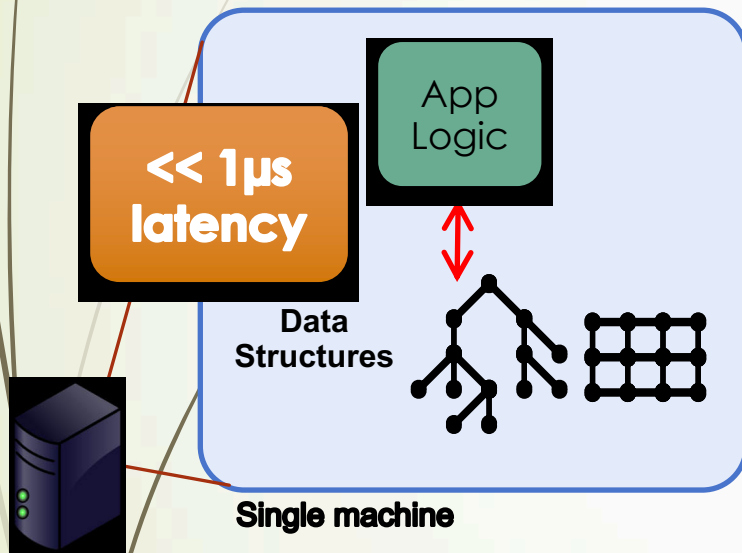
Data Center Traffic Growth



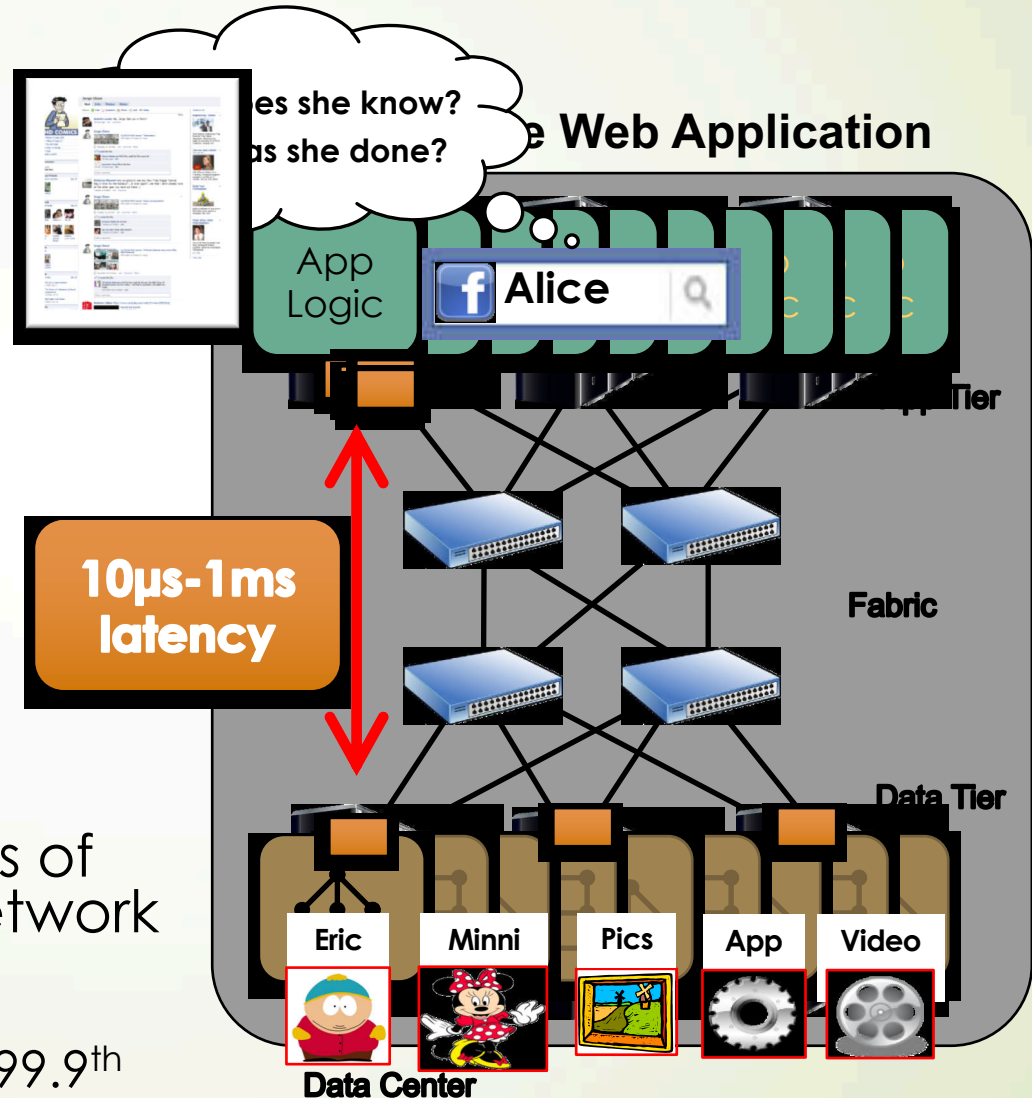
Source: "Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network", SIGCOMM 2015.

In Fact, Data Center Networks are Quite Unique!

Traditional Application



- 1 user request → 1000s of messages over DC network
- ms of latency matter
- Even at the tail (e.g., 99.9th percentile)



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